14. Exploring Pro-Code Extension of a Zoho Creator Application Using Zoho Catalyst / Custom Functions

Aim

To study how a Zoho Creator application can be extended with pro-code capabilities by using custom Deluge functions and by calling a serverless function built on Zoho Catalyst.

Theory

- **Custom Deluge function:** reusable logic written once in Deluge and called from workflows, buttons and other scripts.
- **Zoho Catalyst:** a serverless platform (Node.js, Java, Python) for heavy processing or custom back ends, called from Creator through an authorised connection and the invokeurl task.

Procedure

These steps describe how a Zoho Creator application is extended with pro-code. The code shown is illustrative.

1. Open the application, go to the functions option and see how a custom function is created with its name, return type and parameters.
2. Study how the Deluge logic is written, tested in the editor and saved. An illustrative function that returns how many days an item has stayed unclaimed:

```
int daysUnclaimed(date foundDate)
{
    return foundDate.daysBetween(zoho.currentdate);
}
```

3. Study how the function is called from a form workflow or a report action: d = thisapp.daysUnclaimed(input.Date_Found); info d;
4. Study how a Catalyst project is created and a small Node.js Basic I/O function is deployed, and how a connection to it is created from Zoho Creator.
5. Study how the Catalyst endpoint is called with invokeurl from a Creator script:

```
resp = invokeurl
[
    url :"<Catalyst function endpoint>"
    type :GET
    connection :"catalyst_conn"
];
info resp;
```

For the lost and found application, a pro-code extension would be considered for matching lost and found items by text similarity, for reading an identity card by OCR, or for sending messages through an external gateway.

Result

Custom Deluge functions and the use of a Zoho Catalyst serverless function through invokeurl were studied, and the points at which the lost and found application would need such a pro-code extension were identified.